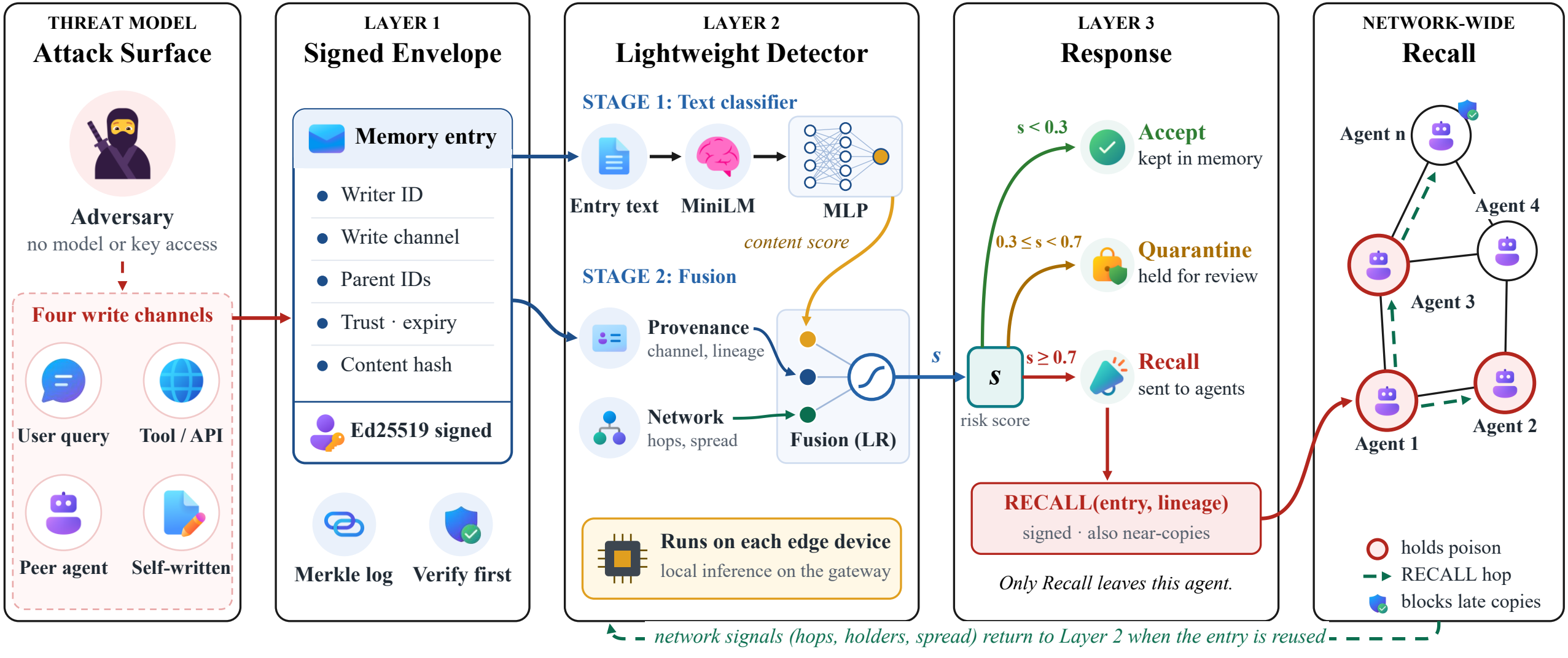

THREAT MODEL
Attack Surface
Adversary
no model or key access
Four write channels
User query
Tool / API
Peer agent
Self-written
LAYER 1
Signed Envelope
Memory entry
Writer ID
Write channel
Parent IDs
Trust · expiry
Content hash
Ed25519 signed
Merkle log
Verify first
LAYER 2
Lightweight Detector
STAGE 1: Text classifier
Entry text
MiniLM
MLP
content score
STAGE 2: Fusion
Provenance
channel, lineage
Network
hops, spread
Fusion (LR)
Runs on each edge device
local inference on the gateway
s
LAYER 3
Response
s < 0.3
Accept
kept in memory
0.3 ≤ s < 0.7
Quarantine
held for review
s ≥ 0.7
Recall
sent to agents
risk score
RECALL(entry, lineage)
signed · also near-copies
Only Recall leaves this agent.
NETWORK-WIDE
Recall
Agent n
Agent 4
Agent 3
Agent 1
Agent 2
holds poison
RECALL hop
blocks late copies
network signals (hops, holders, spread) return to Layer 2 when the entry is reused